

Charalambos Mitropoulos

✉ charalambos.mitropoulos@gmail.com

🔗 chamitro.github.io

in Charalambos Mitropoulos

🔗 chamitro

Research Interests

My research interests lie in the area of smart contracts, program analysis, and software testing.

Education

PhD in Computer Science – Technical University of Crete

Sept 2020 – Today

Supervisor: Prof. S. Ioannidis

Indicative courses and grades:

- Advanced Computer Architecture: 10/10
- Computer and Systems Security: 7/10

MSc in Information Systems – Athens University of Economics and Business

Sept 2017 – May 2019

GPA: 8/10

Thesis: Employing Different Program Analysis Methods to Study Bug Evolution

Supervisor: Prof. D. Spinellis

Indicative courses and grades:

- Software Reliability and Quality Assurance: 8/10
- Computer Security: 8/10
- Cryptography: 9/10

BSc in Informatics and Telecommunications – National and Kapodistrian University of Athens

Sept 2011 – Sept 2017

GPA: 7/10

Thesis: Local Search for Optimization Problems

Supervisor: Prof. V. Zissimopoulos

Indicative courses and grades:

- Operating Systems: 8/10
- Algorithms and Complexity: 8/10
- Systems Programming: 7/10

High School Diploma

Sept 2011

GPA: 18.5/20

Awards and Scholarships

Alpha Bank's Scholarship for First-Year Students

2012-2013

A scholarship provided by the bank to students who have GPA above 7.5/10

SIGSOFT CAPS

August 2019

Student Travel Support and Childcare Assistance at Conferences. The grant was given by the organizing committee of the ESEC/FSE '19.

Experience

Research Associate

January 2019 – Today

Business Analytics Lab ([Balab](#) [🔗](#)), Athens University of Economics and Business

Research on software security and testing

Web Development, IT support

June 2016 – March 2026

Development and support of the [Dikotylon](#) [🔗](#) website.

Technologies: HTML, JavaScript

Researcher at Foundation of Research and Technology-Hellas

August 2020 – December

Participated in European funded programmes :

2021

- ASCAPE
- SPIDER 5G CYBER RANGE

Researcher at Technical University of Crete

December 2021 -

Participated in European funded programmes :

December 2025

- CyberSecPro
- Sentinel
- SecOpera
- Cybersecdome

Partner at DIENEKES

January 2026 - Today

Participated in European funded programmes :

- SYNAPSE (EU Award Number: 101120853)

Publications

- **Charalambos Mitropoulos**, Maria Kechagia, Chrysostomos Maschas, Sotiris Ioannidis, Federica Sarro, and Dimitris Mitropoulos. [Broken agreement: The evolution of Solidity error handling](#) . In *Proceedings of the 18th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM '24)*. Pages 257 - 268. October 2024.
- **Charalambos Mitropoulos**, Thodoris Sotiropoulos, Sotiris Ioannidis, Dimitris Mitropoulos. [Syntax-Aware Mutation for Testing the Solidity Compiler](#) . In *Proceedings of the 28th European Symposium on Research in Computer Security (ESORICS '23)*. Pages 327–347. Springer. September 2023.
- Stefanos Chaliasos, Thodoris Sotiropoulos, Georgios-Petros Drosos, **Charalambos Mitropoulos**, Dimitris Mitropoulos, and Diomidis Spinellis. [Well-typed programs can go wrong: a study of typing-related bugs in JVM compilers](#) . In *Proceedings of the ACM on Programming Languages, Volume 5, Issue OOPSLA (part of SPLASH 2021)*. Article No. 123. Pages 1-30. October 2021.
- **Charalambos Mitropoulos**, “Employing Different Program Analysis Methods to Study Bug Evolution” . In *Proceedings of the 2018 26th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE '19): Student Research Competition (SRC) Track*. 2019. August 2019. Pages 1202–1204.

PC Member

- 2025: ISSTA (Artifact Evaluation Committee)

Teaching Assistant

Computer Security: Computer Security. Technical University of Crete. BSc Course 5th semester. 2022-Today.

Talks

- **Seminar Presentation @AUEB's BALAB (2023):** Syntax-Aware Mutation for Testing the Solidity Compiler
- **PL-Seminar @National Technical University of Athens, Software Engineering Laboratory (2023):** Syntax-Aware Mutation for Testing the Solidity Compiler
- **Seminar Presentation @AUEB's BALAB (2024):** Broken Agreement: The Evolution of Solidity Error Handling

Personal Projects

Testing the Solidity Compiler:

Fuzzing the Solidity (an object-oriented programming language for writing smart contracts) compiler with AFL (American Fuzzy Lop). Notably, the upstream developers have provided instant bug-fixes releases after approving my patches. Some of these patches are the following:

- [ICE on missing modifier identifier 7550](#) 
- [ICE on array index with ABIEncoderV2 7549](#) 

- ICE when assigning to calldata structs and static arrays using using inline assembly + FixedPointType not Implemented error 11680 [↗](#)
- ICE in CompilerUtils::convertType when converting a calldata slice of bytes or string 11677 [↗](#)

Technical Skills

Programming Languages: C, C++, Java, SQL, Python, PHP, Prolog, Scala, HTML, JavaScript, Bash scripting

Tools and Editors: Tools and Editors: Vim, IntelliJ, Eclipse, Netbeans, Hadoop, MySQL, MySQL Workbench

Documentation and Preparation Systems: LaTeX, Word, Excel, Powerpoint

Foreign Languages

English: Certificate of Proficiency in English, University of Michigan.

German: Goethe – Zertificat A